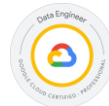


David Neish



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SUMMARY:

Solutions Architect with 4.5+ years building distributed systems in finance, now focused on applied AI and LLM infrastructure. I build software systems that people use, and businesses profit from. Pursuing an M.S. in Human-Computer Interaction to advance human-centered agent design and multimodal feedback architectures.

SKILLS

Languages & Tools: Java, Maven, Python, Pytorch, SQL, C++, Git, gRPC, REST, NATS, Kafka

AI/ML Frameworks: Langchain, LangGraph, LlamaIndex, trl, NVIDIA NIM, Nemo, TensorRT

Cloud & DevOps: AWS, EKS, GCP, Google Cloud Run, Azure AI Foundry, Vertex AI, Docker, Kubernetes, GitHub Actions, NVIDIA GPU Operator, MIG, Infiniband

Data & Databases: PostgreSQL, MySQL, Redis, BigQuery, Oracle DB, NoSQL, Hazelcast, Ignite

Finance-Specific: ISO 8583, ISO 20022, Protobuf

EXPERIENCE

Machine Learning Software Engineer | MASTERCARD | AUGUST 2021 – PRESENT

- Designed and maintained distributed, event-driven applications using NATS for real-time pub/sub messaging, supporting mission-critical financial workflows processing 10M+ daily transactions and contributing to the system reliability needed for Mastercard's transition toward real-time clearing and faster settlement cycles.
- Architected and optimized a Databricks + LangChain retrieval pipeline (BM25, metadata tagging, Llama-70B), improving task success rate from 0% to 92% after redesigning retrieval and embedding architecture. Enabled 20+ global sales specialists to access case studies faster, accelerating campaign delivery by 60% and improving credit-card activation targeting.
- Architected an AI-powered Python web app using NiceGui and GPT-4 to reduce language translation time for PowerPoint, Docs, and PDFs by 95%, with an error rate under 5%. Realized cost savings of up to 99% compared to traditional agency-based translations.
- Built a segment-level feedback system for LLM completions with edit/approve/refine actions, converting user interactions into high-quality JSONL training data for reusable pipelines.
- Developed a containerized testing framework using Testcontainers to reproduce distributed-system behaviors such as network partitions and split-brain cache states across 4 controlled scenarios. Enabled earlier fault detection and improved confidence across teams responsible for high-volume event-driven services.
- Served as an AI adoption advisor, briefing executives on coding agents, LLM strategy, and industry direction. Built and manage a 37-model governance registry that standardizes evaluation, mitigates risk, and accelerates enterprise AI integration.
- Built an automated CI/CD pipeline for registering Protobuf schemas, replacing a manual PR process with auto-approvals via Bitbucket's API. Integrated Salesforce Maven Plugin to verify backward compatibility, saving 250+ hours quarterly and enforcing consistent, production-safe schema updates.

SOFTWARE ENGINEER INTERN | FIDELITY INVESTMENTS | JUNE 2020 – AUGUST 2020

- Documented cloud adoption patterns and developed an AWS S3 onboarding guide, leveraging CLI and SDK workflows to simplify file transfers and storage setup for new teams.
- Built a Java Spring Boot utility app that automated common S3 bucket operations, reducing manual steps and easing developer onboarding.

Education

SUNY Fredonia | B.S. Computer Science | May 2021

Indiana University | M.S. Human-Computer Interaction | May 2027 |

Certifications

AWS Certified Solutions Architect – Associate | Azure AI Fundamentals | GCP Professional Data Engineer | NVIDIA Generative AI with Diffusion Models | NVIDIA AI & Infrastructure and Operations Fundamentals | NVIDIA Fundamentals of Deep Learning

Applied AI Projects & Agentic Systems

Built a set of **Financial Services modeling notebooks** covering: fine-tuning Gemma 270M on the Financial PhraseBank dataset, synthetic data generation using a low-resource language dataset, customer conversion modeling with XGBoost on bank marketing data, and simple probabilistic stock forecasting with Chronos.

Architected **Risky**, an open-source LLM-powered financial advisor using **NVIDIA NIM 405B** and **LlamaIndex** to deliver personalized investment recommendations. Integrated real-time market data, vectorized financial docs, and agentic planning workflows. Designed as a reference architecture for **enterprise-grade, explainable GenAI systems**, supporting dynamic portfolio analysis and modular strategy generation.

Built **Infera**, an AI-driven platform that automates financial reporting, leadership profiling, and market research. Uses **LangGraph** to structure agent workflows with **GPT-4**, Wikipedia, Yahoo Finance, and Google Search. Delivered interactive radar charts and AI-generated rankings to help analysts assess market positioning in seconds, not hours.

Architected a multilingual **document and voice translation system** using **GPT-4** and **Whisper**, deployed on **Google Cloud Run** with segment-level control and human-in-the-loop editing. Automated CI/CD with GitHub Actions for rapid iteration. Achieved **95% reduction in translation time** and **99% cost savings**, delivering a scalable, production-grade alternative to agency-based workflows.

Built “**HoodMatcher**,” a real-estate recommendation engine implementing a full Retrieval-Augmented Generation (RAG) architecture with GPT-4, FAISS semantic search, Zillow data and St Louis crime data. Produced data-grounded generative narratives that combine housing trends and amenities, visualized through Vertex AI Imagen and Streamlit for an interactive, multimodal experience.